

Paper 4.25 - Toolkit for Boundary Repair

CQE-paper-04.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-04.25.md

Paper 4.25 - Toolkit for Boundary Repair

Purpose

Paper 4.25 describes the review tools for boundary repair. The tools help a reader construct and inspect repair rows. They do not add claims beyond the Paper 4 theorem.

Review Objects

The toolkit works with:

correction state	= Paper 2 residue state
axis_sheet coordinate	= Paper 3 registration
repair row	= typed boundary constraint
next legal route	= route allowed to consume the constraint

Digital Toolkit

Primary executable files:

```
production/formal-papers/CQE-paper-04/verify_boundary_repair.py
production/formal-papers/CQE-paper-04/boundary_repair_receipt.json
```

Additional package evidence named by the dyad review:

```
D:\CQE_CMPLX\CMPLX-R30-main\PROOF\src\lattice_forge\binary_boundary_adapter.py
D:\CQE_CMPLX\CMPLX-R30-main\PROOF\src\lattice_forge\transport_obligations.py
```

The verifier checks that Paper 2 correction states are consumed, Paper 3 coordinates are preserved, required fields are present, constraints are not proofs, next legal routes are nonempty, repair is idempotent, and untyped failures are rejected.

Analog Toolkit

A physical reconstruction requires:

- one correction row,
- one coordinate lookup,
- one repair-row template,
- one next-route label,
- one receipt line.

Procedure:

```
copy the Paper 2 correction state
copy the Paper 3 axis/sheet coordinate
write the reason: correction fired
write status: constraint
write at least one next legal route
write source paper and target paper
repeat the repair operation and confirm the row does not change
```

Hidden-Guess Diagnostics

When training mode is enabled, diagnostic review should ask for a prediction before revealing the receipt:

What fields are required for a repaired boundary row?
Does a repaired row count as proof?
Which two states produce repair rows?
Is {"status":"failed"} repairable?

Expected answers:

state, axis_sheet, reason, status, next_legal_routes, source_paper, target_paper
no
(0,1,0) and (1,1,0)
no

Boundary

Paper 4.25 is a toolkit supplement. If a new adapter, route, or domain example is found, it must pass Paper 4.50's claim contract before it changes the scientific paper.